Affective Mirage? Diagnosing Visual Grounding in Emotion Recognition with Vision Language Models

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***Abstract*— Vision-language models (VLMs) have shown remarkable visual understanding capabilities, yet recent studies reveal a prevalent MIRAGE phenomenon: *models may produce plausible predictions without faithfully relying on the underlying visual evidence*. We investigate whether such behavior also exists in the visual emotion recognition task, which is inherently subjective and often relies on subtle and ambiguous visual cues, potentially making the task particularly susceptible to MIRAGE. Thus, we thoroughly examine open-source, closed-source, and domain-specific VLMs through a novel three-level diagnosis framework, named Affective MIRAGE. First, general mirage removes visual evidence to examine model behavior without reliable visual input. Second, causal mirage uses controlled intervention to examine how different visual evidence causally influences model predictions, together with mechanism-level analysis of visual grounding. Third, certainty mirage examines whether model uncertainty follows human disagreement on ambiguous emotions. Our findings reveal three characteristic behaviors of current VLMs in emotion recognition: 1) without visual evidence, all evaluated VLMs maintain a nearly 100% mirage rate under the basic prompt without an abstention option, while prediction collapse exists toward label priors; even when an explicit *cannot determine* option is offered, only closed-source models abstain reliably, 2) the causal dominance of subject evidence is over scene context under counterfactual interventions, and 3) model uncertainty partially tracks human affective ambiguity, yet overconfidence persists on images that humans find ambiguous. To our knowledge, this is the first systematic diagnostic study of MIRAGE behavior in VLM-based visual emotion recognition, identifying limitations beyond benchmark accuracy and offering new insights for future affective VLM evaluation and benchmark design. The code will be available at https://affective-mirage.github.io/Affective_Mirage/.**

I. Introduction

Understanding human emotion from visual input is fundamental to socially aware artificial intelligence [1], [2]. Recent vision-language models (VLMs), including both general-purpose [3]–[5] and affect-specialized models [6], [7], have demonstrated strong benchmark performance on visual emotion recognition. However, benchmark accuracy alone does not reveal how these models reach an affective perception [8], motivating a deeper investigation into their prediction behavior.

This concern is echoed by the recent *MIRAGE* phenomenon in VLMs [9], where models may produce seemingly competent predictions by implicitly imagining nonexistent visual content to answer the user query, rather than grounding their responses in the actual visual evidence. Existing studies [10]–[14] mainly examine conventional visual tasks in which relevant evidence is associated with identifiable objects or regions. Emotion recognition presents a more complex setting. Affective evidence can be distributed across facial expressions, body posture, nearby people, and the context of the scene [15]–[18], so affect at the scene-level can be incorrectly attributed to the target person [19]. Moreover, emotion labels are inherently subjective, and ambiguous expressions often produce substantial disagreement among human annotators [20].

We therefore **investigate whether MIRAGE-like behavior also emerges in visual emotion recognition and how it differs from that observed in conventional visual tasks**. As illustrated in Fig. 1, we propose Affective MIRAGE to diagnose this mechanism across three levels. **General mirage** asks what VLMs predict when visual evidence is unavailable in Diagnosis 1 (D1). **Causal mirage**[1] examines whether surrounding scenes substitute for removed person evidence in Diagnosis 2 (D2) and whether person or scene evidence governs predictions when the two conflict in Diagnosis 3 (D3). **Certainty mirage** investigates whether model uncertainty aligns with human disagreement on ambiguous emotions in Diagnosis 4 (D4). Together, these diagnoses examine not only whether VLMs use visual evidence, but also which evidence they rely on and whether their certainty reflects the ambiguity of that evidence.

Across the three-level diagnostic framework, we conduct extensive experiments with seven state-of-the-art VLMs on four emotion recognition datasets. Three main findings emerge. First, MIRAGE-like behavior also exists in visual emotion recognition, but differs from conventional visual tasks: when visual evidence is unavailable, VLMs rarely proactively signal its absence unless an abstention option is explicitly offered, similar to recent observations that models often fail to recognize when not to respond [21], and their predictions exhibit clear **prediction collapse** in VLMs [22], [23], with distinct patterns toward the dominant emotion categories; even with the option, abstention is not reliably evidence-aware for open-source models. Second, controlled removal and counterfactual recombination reveal a stronger causal influence of subject evidence than scene context on VLM predictions, supported by mechanism-level analyses. Third, model uncertainty partially tracks human affective ambiguity and is reasonably calibrated overall, but remains overconfident specifically on images that humans find am-

[1]We use *causal* in the intervention-based sense: changing subject or scene evidence while holding the other fixed and measuring its effect on model predictions. The scope is limited to model-level effects, not the causal generation of human emotion.

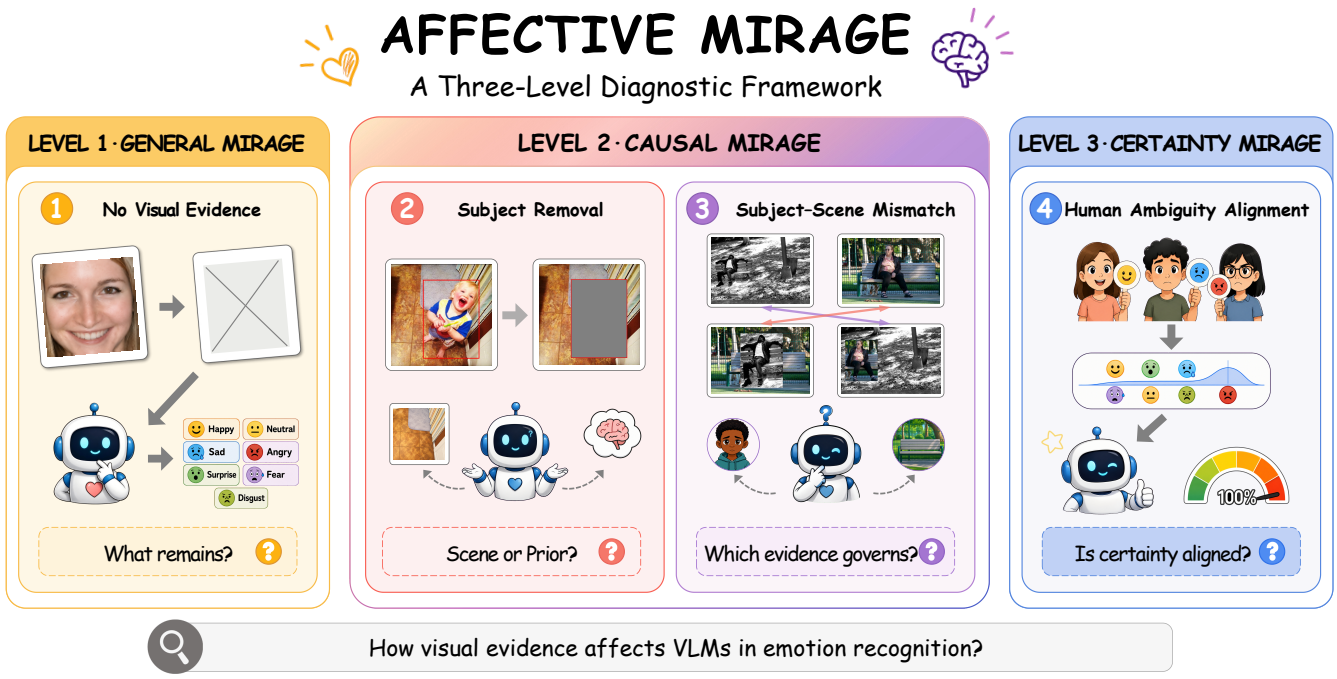


Fig. 1. **Overview of our three-level diagnostic framework for affective mirage.** Level 1 examines predictions without visual evidence (D1), Level 2 probes causal reliance on subjects and scenes (D2 and D3), and Level 3 evaluates certainty alignment with human ambiguity (D4). To the best of our knowledge, this is the first work to systematically diagnose affective MIRAGE in VLM-based emotion recognition.

biguous. Together, these findings reveal underexplored operating patterns of affective VLMs and provide implications for future model and benchmark design, as discussed in Sec. V.

Our main contributions are summarized as follows:

- We propose a **three-level diagnostic framework** comprising general, causal, and certainty mirage to examine visual dependence, the causal influence of visual evidence, and uncertainty alignment. To our knowledge, this is the first systematic study of MIRAGE-like behavior in visual emotion recognition.
- Through controlled interventions and mechanism-level analyses, we uncover prediction collapse, the causal dominance of subject evidence over scene context, and partial uncertainty alignment with ambiguity-specific overconfidence.
- We show that **benchmark accuracy alone is insufficient for evaluating affective VLMs**, translate these findings into guidance for affective VLM research, including no-vision controls with an explicit abstention option, generality tests for specialized models, and multi-annotator benchmark design.

II. Related Works

A. Visual Emotion Recognition with VLMs

General VLMs enable visual emotion recognition through transferable image-text representations and instruction following [24]–[26]. Building on this paradigm, affective VLMs introduce emotion-specific multimodal adaptation [6], [7], [27], while recent benchmarks extend evaluation from emotion classification to affective understanding, reasoning, and generation [28]–[30]. Visual emotion can arise from both the target person and the surrounding context [31]–[33]. Recent studies investigate contextual and facial biases in VLM-based emotion recognition [34], [35]. However, existing works primarily characterize and mitigate specific contextual or facial biases. In contrast, we combine subject–scene interventions with mechanism-level analysis to distinguish evidence-driven predictions from label-prior collapse and identify which visual evidence governs model predictions.

B. MIRAGE and Visual Grounding Diagnostics

Recent studies question whether strong VLM performance necessarily reflects genuine visual understanding. MIRAGE [9] shows that VLMs can generate image-grounded answers and retain nontrivial benchmark performance even when images are absent. Related work examines language priors through staged capability tests and counterfactual images [10], while visual counterfacts have been used to separate reliance on image evidence from stored knowledge [36]. Mechanistic studies further probe where visual information enters the model and use internal interventions to examine its causal influence on generated outputs [37], [38]. EmotionHallucer [39] benchmarks emotion hallucinations in multimodal models through adversarial question answering. ELENA [35] uses attention analysis to reveal persistent facial bias in embodied emotion recognition. These analyses focus on general visual reasoning or particular failures in affective reasoning. In contrast, our work addresses the distributed and ambiguous nature of affective evidence by jointly diagnosing whether emotion predictions depend on vision, how subject and scene evidence influence them, and whether model

uncertainty reflects human disagreement.

C. Ambiguity in Emotion Recognition

Emotion labels are inherently ambiguous, as the same facial expression may admit multiple human interpretations. This ambiguity is reflected in crowd-annotated datasets and emotion-distribution formulations that preserve disagreement beyond a single hard label [40]–[42]. Prior work addresses such ambiguity through distribution learning, label-correlation modeling, latent ambiguity mining, and uncertainty estimation [43]–[46]. [47] also studies ambiguous emotion recognition with LLMs in text. This work uses human disagreement as a diagnostic signal to test whether pretrained VLM uncertainty aligns with human affective ambiguity beyond majority-label recognition accuracy.

III. Diagnostic Framework

A. Task Formulation

Emotion recognition with a VLM is defined as follows: given a sample i with visual modality V_i and input prompt T, the model predicts an emotion label $y \in \mathcal{Y}$,

$$y^* = \arg\max_{y \in \mathcal{Y}} p_{\mathcal{M}}(y \mid V_i, T), \tag{1}$$

where $p_{\mathcal{M}}(y \mid V_i, t)$ denotes the label-level prediction for class y under a specific inference scheme, and the VLM is parameterized by $\mathcal{M}$.

B. Diagnosis 1: General Mirage in Visual Absence

Diagnosis 1 examines how VLMs predict emotions when visual evidence is completely unavailable. For each facial image, we query the model under four stimulus conditions: *image_present*, which uses the original face; *blank*, which replaces it with a fixed 224×224 middle gray image; *noise*, which uses a Gaussian noise image deterministically generated from the sampling seed; *blank_mirage_hint*, which uses the blank image together with misleading metadata stating that an image is attached. These conditions progressively remove visual evidence while testing whether either visual placeholders or an image presence cue can induce affective predictions. Each condition is queried with the *basic* prompt, which lists only the seven emotions, and with the *abstention-option* (AO) prompt, which only appends *cannot determine* to this list; AO on *image_present* additionally measures false abstention on real faces. More details are in Supplementary A (SM A).

Under conditions without visual evidence, ideally, we define **any valid emotion prediction as a mirage** , and an *abstention* represents an evidence-aware response from an ideal VLM, where the model explicitly recognizes that the visual information is unavailable. Some VLMs, especially specialized models, fail to parse or execute the task without visual input, which we define as a *parsing failure*. In this work, for simplicity, we consider both any valid emotion prediction and parsing failures as mirage responses as they both indicate a failure to appropriately handle the absence of visual evidence. The corresponding **Mirage Rate** R_{mirage} is the proportion of inference runs that still yield a valid emotion prediction or parsing failure.

To reliably assess the robustness of MIRAGE behavior to stochastic variation, in addition to a single-round verification, we also repeat each no-visual-evidence condition for S (50 in practice) independent runs, using a different random seed for each run. Each seed determines both the Gaussian noise realization, when applicable, and the sampling pattern (temperature=1.0).

As all evaluated VLMs exhibit prediction collapse under a fixed seed (Sec. IV-B), we further examine whether this behavior persists across random seeds. We define the collapse rate as:

$$R_{\text{collapse}} = \frac{1}{S} \sum_{s=1}^{S} \mathbb{I}\left[\hat{y}_s = y_{\text{collapse}}\right], \qquad S = 50, \tag{2}$$

where y_{collapse} denotes the collapse label identified under the fixed-seed setting, which is *neutral* for most models.

The R_{mirage} measures failure to produce an explicit abstention under the tested prompt, whereas R_{collapse} measures the stochastic stability of predictions toward a predefined collapse label.

C. Diagnosis 2: Causal Mirage in Subject Removal

Diagnosis 2 tests whether VLM's emotion prediction depends on visual evidence from the designated subject. Since visual emotion can be informed by both the subject and scene context [31]–[33], D2–D3 intervene on these two sources. For D2, we replace the subject bounding box b with a uniform gray region to obtain $\widetilde{V}_i^b$ and compare

$$p_{\mathcal{M}}(y \mid V_i, T) \quad \text{and} \quad p_{\mathcal{M}}(y \mid \widetilde{V}_i^b, T). \tag{3}$$

Systematic prediction changes reflect sensitivity to the subject-region evidence. Besides, Matched-mask and marker controls are provided in SM A and SM E. Intervention effects are assessed through recognition performance and prediction distributions, and additional analyses are provided in SM A and SM D. In addition, the removal with random RGB texture is repeated in the same subject box to test the dependence on fill appearance (See SM E).

D. Diagnosis 3: Causal Mirage in Subject–Scene Mismatch

Because emotion may depend on multiple visual evidence sources, we perform an evidence-swapping diagnosis that independently swaps the subject region and scene context.

First, a sample i and a sample j are selected according to a pre-specified pairing rule (see SM A) such that their emotion annotations provide conflicting evidence. We conceptually decompose each visual input into the bounding-box subject region and the remaining scene context.

$$V_i = (S_i, C_i), \tag{4}$$

where S_i denotes the rectangular visual region inside the subject bounding box and C_i denotes the remaining visual content. Then, a crossed 2×2 source design is built:

	C_i	C_j
S_i	(S_i, C_i)	(S_i, C_j)
S_j	(S_j, C_i)	(S_j, C_j)

(5)

TABLE I

THE MAIN RESULTS OF D1–D4. ACC@1 IS REPORTED FOR RAF-DB AND CAER AND HIT@1 FOR EMOTIC. FOR D1–D3, V_i IS THE ORIGINAL VISUAL INPUT, $\widetilde{V}_i^b$ REMOVES ITS SUBJECT, (S_i, C_j) COMBINES THE CORRECT SUBJECT WITH A MISMATCHED SCENE, AND (S_j, C_i) THE CORRECT SCENE WITH A MISMATCHED SUBJECT. R_{MIRAGE}: MIRAGE RATE. SS AND MS DENOTE SINGLE- AND MULTI- SEED SETTINGS, RESPECTIVELY. ALL RESULTS USE THE BASIC PROMPT EXCEPT THE AO ROW, WHICH ADDS A *cannot determine* OPTION; FOR CLOSED-SOURCE MODELS, THE THIRD NO-VISION INPUT SENDS NO IMAGE (TRUE-ABSENT) INSTEAD OF HINT. PARENTHESES SHOW CHANGES FROM V_i. FOR D4, ρ MEASURES ALIGNMENT BETWEEN MODEL AND HUMAN ENTROPY, WHILE p_{top1} DENOTES MEAN PREDICTION CONFIDENCE.

Diagnosis	Dataset	Input / metric	GPT-5.6 Luna [48]	Gemini-3.5 Flash-Lite [49]	Qwen3.5 -9B [3]	InternVL3.5 -8B [4]	Ministral 3 8B [5]	Emotion -LLaMA* [6]	AffectGPT [7]
Diagnosis1	RAF-DB [41]	V_i	0.737	0.646	0.692	0.762	0.546	0.747	0.182
		No visual Acc	0.222 (−0.515)	0.222 (−0.424)	0.222 (−0.470)	0.222 (−0.540)	0.222 (−0.324)	0.222 (−0.525)	0.210 (−0.028)
		R_{mirage} (SS)§	100%	100%	100%	100%	100%	100%	100%
		R_{mirage} (MS)§ (blank/noise/hint)	100%/100%/100%	100%/100%/100%	**98%/98%/92%**	100%/100%/100%	**94%/88%/94%**	100%/100%/100%	**98%**/100%/100%
		R_{mirage} AO (MS)§ (blank/noise/hint)	0%/0%/0%	0%/0%/0%	28%/14%/40%	**86%**/2%/78%	4%/0%/22%	100%/100%/100%	–
Diagnosis2 & Diagnosis3	EMOTIC [32]	V_i	0.819	0.868	0.755	0.842	0.754	0.187	0.553
		$\widetilde{V}_i^b$	0.437 (−0.381)	0.774 (−0.094)	0.487 (−0.269)	0.695 (−0.147)	0.563 (−0.191)	0.156 (−0.030)	0.550 (−0.003)
		(S_i, C_j)	0.670 (−0.148)	0.817 (−0.051)	0.493 (−0.262)	0.675 (−0.166)	0.513 (−0.240)	0.157 (−0.030)	0.559 (+0.006)
		(S_j, C_i)	0.225 (−0.593)	0.353 (−0.515)	0.246 (−0.509)	0.324 (−0.518)	0.385 (−0.369)	0.144 (−0.043)	0.549 (−0.004)
	CAER [33]	V_i	0.528	0.540	0.530	0.416	0.456	0.220	0.347
		$\widetilde{V}_i^b$	0.351 (−0.177)	0.351 (−0.189)	0.355 (−0.175)	0.346 (−0.069)	0.344 (−0.112)	0.127 (−0.093)	0.344 (−0.004)
		(S_i, C_j)	0.501 (−0.026)	0.532 (−0.007)	0.513 (−0.017)	0.412 (−0.004)	0.445 (−0.011)	0.219 (−0.001)	0.348 (+0.001)
		(S_j, C_i)	0.146 (−0.382)	0.191 (−0.349)	0.180 (−0.350)	0.104 (−0.311)	0.187 (−0.268)	0.107 (−0.113)	0.343 (−0.004)
Diagnosis4 †	FER+ [40]	ρ_{all}	0.403	0.357	0.601	0.548	0.233	0.365	-0.245
		ρ_{correct}	0.453	0.406	0.575	0.560	0.229	0.541	-0.077
		p_{top1}	0.681	0.734	0.855	0.843	0.822	0.547	0.992

§ MIRAGE includes *any emotion prediction* or *parsing failure*; only *abstention* is non-MIRAGE. SM C.4 reports the component rates separately. AffectGPT AO results are pending.
* Emotion-LLaMA uses the EMOTIC target image or CAER's first sampled frame, with unavailable auxiliary inputs zero-filled.
† For D4, closed-source values (gray) are self-reported confidence rather than model posteriors, because their output heads are inaccessible; they are not compared with open-source models.

The diagonal combinations represent the original source pairings, whereas the off-diagonal combinations swap the subject and contextual sources between samples i and j.

To construct each crossed input, the original subject region is directly resized to the destination box using bicubic interpolation and pasted onto the target canvas, preserving the source region without cropping or padding. The same procedure is applied to both crossed conditions.

This crossed design further allows us to characterize which evidence dominates the resulting prediction. Let $\mathcal{E}$ denote the set of candidate visual evidence sources which may causally affect the prediction, and let $\mathcal{Y}_{e,i}$ denote the emotion annotations associated with the visual evidence source $e \in \mathcal{E}$ for sample i. Importantly, $Y_{e,i}$ can serve as an annotation proxy for source e, rather than necessarily constituting an independent ground-truth annotation of that source. Considering that the predicted label y_i may fall into the annotation sets of multiple visual evidence sources, we assign y_i to the source e when it belongs only to $\mathcal{Y}_{e,i}$ and not to any other annotation sets:

$$a_{i,e} = \mathbb{I}\left[y_i^* \in \mathcal{Y}_{e,i} \setminus \bigcup_{k \in \mathcal{E},\, k \neq e} \mathcal{Y}_{k,i} \right]. \tag{6}$$

Then, we define the **Causal Evidence Dominance (CED) Score** of e as

$$D_e = \frac{1}{N}\sum_{i=1}^{N} a_{i,e} - \max_{k \in \mathcal{E},\, k \neq e} \frac{1}{N}\sum_{i=1}^{N} a_{i,k}, \tag{7}$$

where the first term indicates the number of samples aligning with e, and thus the second term represents its strongest competing visual evidence source. Accordingly, a positive D_e indicates that e dominates all competing visual evidence sources in directing the model prediction, with larger values indicating stronger dominance over its strongest competitor. The dominant visual evidence source is therefore

$$e^\star = \arg\max_{e \in \mathcal{E}} \frac{1}{N}\sum_{i=1}^{N} a_{i,e}. \tag{8}$$

Predictions that cannot be uniquely attributed to any source are treated as unresolved, with rate

$$F_U = 1 - \frac{1}{N}\sum_{i=1}^{N}\sum_{e \in \mathcal{E}} a_{i,e}. \tag{9}$$

In our subject-context intervention, $\mathcal{E} = \{\mathrm{S}, \mathrm{C}\}$, where S and C denote the subject and scene-context evidence sources, respectively. In this context, for the crossed input (S_j, C_i), $\mathcal{Y}_{s,i}$ uses the subject annotation of sample j, while $\mathcal{Y}_{c,i}$ uses the subject annotation of sample i itself, as no independent

scene-only annotation is available. Here, $Y_{c,i}$ is a source-associated proxy rather than a scene-only ground truth; thus, CED measures relative subject–context alignment under counterfactual recombination, not intrinsic scene affect.Since our setting contains only two evidence sources, $D_{\mathrm{C}} = -D_{\mathrm{S}}$. We therefore report only D_{S}.

Similar to Sec. III-C, we additionally conduct a marker-free subject-region replacement to disentangle the effect of visual manipulation from explicit attention guidance. See SM A for details. In addition, the experiment is conducted, where samples with the same or a different emotion label are used to replace the subject or scene. For each of (S_j, C_i) and (S_i, C_j), we count how often the predicted label differs from that for V_i and compare the two category conditions. See SM E.

E. Diagnosis 4: Certainty Mirage in Affective Ambiguity

The final diagnosis examines whether VLMs exhibit affective uncertainty that reflects ambiguity in human emotion judgments.

For open-source models, we use teacher forcing to obtain the log-likelihood $\ell_{i,c}$ of each candidate emotion label y_c given the image and prompt x_i. These scores are normalized over the candidate emotion space to form the model class distribution $\mathbf{q}_i$:

$$\ell_{i,c} = \log p_\theta(y_c \mid x_i), \qquad q_{i,c} = \frac{\exp(\ell_{i,c})}{\sum_{c' \in \mathcal{Y}} \exp(\ell_{i,c'})},$$
$$H_{\mathrm{M}}(i) = -\sum_{c \in \mathcal{Y}} q_{i,c} \log_2 q_{i,c}. \tag{10}$$

Here, $q_{i,c}$ denotes the normalized model probability assigned to class c. $H_{\mathrm{M}}(i)$ is the entropy of the model prediction, with higher values indicating greater affective uncertainty. We additionally report $p_{\mathrm{top1}}(i)$, the probability assigned to the predicted class, as a measure of prediction confidence. Full details of the candidate scoring procedure are provided in the SM A.

Closed-source models are analyzed separately using their self-reported structured confidence distributions, since they do not expose hidden states or vocabulary logits. These results are reported for reference only and are not used to support our main conclusions.

To obtain the human reference, we normalize the 10-annotator votes for each image into a distribution $\mathbf{h}_i$ and compute its entropy $H_{\mathrm{H}}(i) = -\sum_c h_{i,c} \log_2 h_{i,c}$. A larger $H_{\mathrm{H}}(i)$ represents greater disagreement among human annotators. We measure human-aligned affective uncertainty using the Spearman correlation $\rho_{\mathrm{S}}(H_{\mathrm{H}}, H_{\mathrm{M}})$, which tests whether the model becomes more uncertain on images that humans also perceive as emotionally ambiguous. We further measure calibration against the human majority label with the expected calibration error (ECE) [50] and the Brier score [51], the Jensen–Shannon divergence (JSD) between $\mathbf{q}_i$ and $\mathbf{h}_i$, and the confidence gap $\bar{p}_{\mathrm{top1}} - \mathrm{Acc}$ within human-agreement groups, and apply cross-validated temperature scaling [50] to test whether overconfidence is merely a global scale effect (SM A.4).

IV. Experiments

A. Experimental Setup

Each diagnosis requires different visual evidence and annotations, so we use the benchmark best suited to each objective and use its validation/test sets. Although each diagnosis uses a single dataset or designed datasets, the observed behaviors remain informative for the mirage examined.

For D1, we use RAF-DB [41] (7-class, single-label) and report Acc@1 under *image-present*, *blank*, *noise*, and *mirage-hint* conditions, each under the basic and AO prompts. For D2–D3, we use EMOTIC [32] (26-label, multi-label; Hit@1) and CAER [33] (7-class, single-label; Acc@1). As CAER lacks subject boxes, we obtain pseudo subject regions with YOLOv8 [52]; face/head-region controls using [53], [54] are in SM G. For D4, we use FER+ [40], whose ∼10 crowd annotations per image offer human label distributions, and report Spearman ρ between model and human entropy and calibration metrics (ECE, Brier score, and JSD to the human votes), with majority-label accuracy as a prerequisite. More details are in SM G.

We evaluate three general VLMs, Qwen3.5-9B [3], InternVL3.5-8B [4], and Ministral-3-8B [5]; two affective VLMs, AffectGPT [7] and Emotion-LLaMA [6], as domain-specific representatives; and two closed-source models, GPT-5.6-Luna [48] and Gemini-3.5-Flash-Lite [49]. The input prompt remains consistent within the dataset. Experiments are run on AMD MI250X (D2/D3) and NVIDIA GH200 (D1/D4) GPUs.

B. General Mirage: Label Prior Collapse

As shown in Table I and Table II, most models perform well when visual evidence is available, with five of the seven exceeding 0.64 Acc@1. Once visual input is removed, most models collapse toward predicting *neutral*, reaching an accuracy of 0.222 (the proportion of neutral samples in RAF-DB). **Crucially, the models rarely acknowledge the absence of visual evidence and instead fall back to a dominant label prior.** Under the basic prompt, every model exhibits a **near-100% mirage rate**: 100% for seed 0 and 88–100% across 50 seeds and the three no-vision configurations. When *cannot determine* is explicitly offered (AO, Table II), both closed-source models abstain on every no-vision call with only 2–4% false abstention on real faces, i.e., they can detect the missing evidence but do not proactively report it. Open-source models, however, do not abstain in an evidence-aware way: **InternVL3.5-8B abstains on 98% of noise inputs but still answers on 86% of blank inputs**, mostly with *neutral*, treating a uniform gray image as a neutral face; Emotion-LLaMA and AffectGPT never abstain; and the option raises false abstention on real faces to 12.9% for InternVL3.5-8B and 43.7% for Ministral-3-8B. For these models, the option behaves as one more label to choose rather than a recognition of missing evidence (SM C.4). Unlike the unconstrained guessing reported in conventional MIRAGE settings [9], label-constrained emotion recognition exhibits a distinct failure mode: a concentrated prediction collapse

to the dominant label. We also observe that domain-specific VLMs, such as AffectGPT, contain a different mirage issue (parsing failure). It dips to ∼0.210 because a small fraction of its outputs violate the required format rather than collapsing to *neutral* label; across seeds, AffectGPT frequently produces parsing failures under some conditions. We treat this as an AffectGPT-specific fitting issue that does not affect the overall trend (SM B).

SM C.4 also reports R_{collapse}, the rate of predictions falling into the identified dominant label, *neutral*. Notably, general VLMs largely collapse cleanly to *neutral* across seeds, with both collapse and MIRAGE rates close to 1. In contrast, emotion-specific models retain MIRAGE rates at or near 1 but exhibit substantially lower collapse rates. The causes differ: Emotion-LLaMA predicts other emotion labels, whereas AffectGPT frequently produces parsing failures. We further discuss these distinct behaviors in SM C.

Key Finding 1. Without visual evidence, VLMs rarely proactively signal its absence unless an abstention option is explicitly offered, and their predictions collapse toward a dominant label. With the abstention option, closed-source models abstain reliably, whereas open-source and emotion-specific models do not, exhibiting false abstention rates of 10%–43%. Thus, simply introducing an abstention option does not necessarily improve performance.

TABLE II

D1: BASIC VS. ABSTENTION-OPTION (AO) PROMPT. NO-VISION ENTRIES REPORT R_{MIRAGE} UNDER BASIC → AO, OVER $S = 50$ SEEDS (CLOSED-SOURCE MODELS: 50 CALLS; THEIR THIRD INPUT SENDS NO IMAGE). ACC AND F-ABST (FALSE ABSTENTION) ARE MEASURED ON REAL FACES (*image_present*, FULL RAF-DB TEST SET). COMPONENT RATES ARE IN SM C.4.

Model	Blank	Noise	Hint / No-img	Acc	F-abst
GPT-5.6-Luna	100→0	100→0	100→0	0.737→0.738	2.3%
Gemini-3.5-Flash-Lite	100→0	100→0	100→0	0.646→0.661	3.9%
Qwen3.5-9B	98→28	98→14	92→40	0.692→0.708	0.2%
InternVL3.5-8B	100→**86**	100→2	100→78	0.762→0.685	**12.9%**
Ministral-3-8B	94→4	88→0	94→22	0.546→0.317	**43.7%**
Emotion-LLaMA	100→**100**	100→**100**	100→**100**	0.747→0.745	0.0%
AffectGPT	98→**100**	100→**100**	100→**100**	0.182→0.227	0.0%

TABLE III

D3: CED UNDER SUBJECT REPLACEMENT (S_j, C_i). POSITIVE D_{S} INDICATES SUBJECT-EVIDENCE DOMINANCE; F_{U}: UNRESOLVED RATE.

	EMOTIC		*CAER*	
Model	D_{S}	F_{U}	D_{S}	F_{U}
GPT-5.6 Luna	+32.4%	22.5%	+27.4%	43.4%
Gemini-3.5 Flash-Lite	+19.4%	10.0%	+19.7%	42.2%
Qwen3.5-9B	+27.8%	23.0%	+22.1%	41.8%
InternVL3.5-8B	+16.9%	18.3%	+35.6%	43.5%
Ministral 3 8B	-3.0%	26.0%	+19.4%	43.1%
Emotion-LLaMA	+6.2%	65.1%	+29.9%	48.7%
AffectGPT	-13.3%	3.5%	-22.1%	53.4%

C. Causal Mirage I: Prediction Collapse

Diagnosis 2 examines model behavior when the subject evidence is removed. As shown in Table I, subject removal reduces performance for the five general VLMs by 0.094-0.381 on EMOTIC and 0.069-0.189 on CAER, whereas AffectGPT changes little and Emotion-LLaMA shows only small declines, suggesting different patterns. Furthermore, Macro-F1 degrades more strongly (see SM D).

VLM predictions exhibit clear prediction collapse. Fig. 2 shows that prediction distributions are already skewed under the original input. For instance, despite EMOTIC having 26 labels, the three most frequently predicted labels account for most of GPT-5.6 Luna's predictions. Subject removal further amplifies this pattern, driving predictions toward a smaller set of dominant labels, especially on CAER. Detailed statistics in SM D further show where this collapse occurs: the five general VLMs predominantly shift toward *Disconnection* or *Engagement* on EMOTIC, while all collapse toward *Neutral* on CAER. The CAER face/head control confirms that this is not solely due to large subject regions (SM E). In contrast, Emotion-LLaMA exhibits a distinct pattern from the other VLMs: it collapses to *Affection* on EMOTIC and to *Surprise/Anger* on CAER. Above all, Diagnosis 2 reveals the affective causal mirage: nontrivial Hit@1 or Acc@1 can persist after removing subject evidence because the recognition score is partly sustained by prediction collapse rather than effective substitution by scene evidence.

Controlled experiments further support this interpretation (see SM E). Moving the same gray box away from the subject largely restores performance, ruling out generic occlusion effects. Subject removal also has a stronger effect in single-person images and weakens as more subjects are visible, consistent with compensation from remaining person-level evidence. The drop in recognition performance persists when gray masks are replaced with random texture. CAER predictions also remain concentrated on *netural*, showing that the collapse is not specific to gray masking.

Key Finding 2. VLMs exhibit obvious prediction collapse on emotion prediction tasks, which can be amplified by removing subject-region evidence, though they still achieve nontrivial scores on benchmarks.

D. Causal Mirage II: Subject Evidence Dominance

Diagnosis 3 compares the causal influence of subject and scene evidence using the two crossed conditions (S_i, C_j) and (S_j, C_i). Table I shows larger performance drops under subject replacement than scene context replacement: 0.369-0.593 vs. 0.051-0.262 on EMOTIC, and 0.268-0.382 vs. 0.004-0.026 on CAER. See detailed analysis in SM D.

Table III further identifies the dominance of the visual evidence sources. CED score is positive for four of five general VLMs on EMOTIC and all five on CAER, indicating that when subject and scene evidence conflict, general VLMs usually follow the subject region. In contrast, two affective VLMs are less consistent: Emotion-LLaMA has a positive

CAER score but a high unresolved rate, while AffectGPT follows the retained scene context more often than the inserted subject region on both datasets. Together with Diagnosis 2, these results suggest that scene context rarely substitutes for person-specific affective evidence.

Besides, control analyses support the observed subject dominance (see SM E). First, on CAER, even replacing only the localized face/head region affects predictions substantially more than replacing the remaining context across all five general VLMs. Second, on single-person EMOTIC samples, removing the marker has only a minor effect compared with replacing the subject-region content, indicating that the observed preference is primarily driven by visual evidence within the subject region. Furthermore, on CAER, predictions change more often when the replacement subject has a different emotion label than when it has the same label as the original subject. Comparatively, this difference is not clearly observed for scene replacement. Finally, analyses of resampling effects and prompt variations support the observed regional dependence. See SM E for complete control results for all models.

Then, we examine whether the strong behavioral dependence on subject evidence is reflected in the model's internal states on InternVL3.5-8B, Qwen3.5-9B, and Emotion-LLaMA. We patch visual-token states from the full-input inference into the subject-masked inference and test whether they restore the full-input prediction. For InternVL3.5-8B and Qwen3.5-9B, restoring subject-region-related states recovers the original-input prediction on substantially more samples than restoring states at the remaining image positions (See SM F). Fig. 3 visualizes this effect through prediction restoration maps (PRMs) at 25% decoder depth, where higher values indicate a larger increase in the probability of the full-input prediction after restoring that visual-token state. Thus, the sensitivity to subject evidence is accompanied by a localized internal dependence on subject-associated visual states, providing mechanism-level support for our subject-grounding finding. The algorithm and detailed results are provided in SM F.

Key Finding 3. General VLMs rely predominantly on subject evidence, with scene context failing to substitute when the subject is removed. Affective VLMs show more model-specific source preferences, indicating that emotion specialization does not ensure more consistent causal grounding.

E. Certainty Mirage: Limited Uncertainty Alignment

Diagnosis 4 examines whether model uncertainty increases with human disagreement in emotion judgments. As shown in Table I, four of the five open-source models exhibit significantly positive correlations between model and human entropy. Qwen3.5-9B and InternVL3.5-8B achieve the highest ρ_{all} values of 0.601 and 0.548, respectively, i.e., 89% and 81% of the human reliability ceiling (0.675; split-half with Spearman–Brown correction, SM A.4), indicating

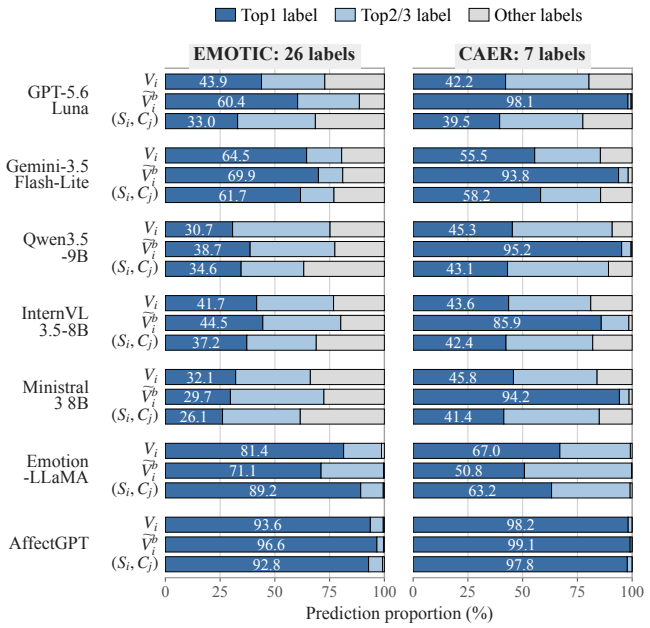


Fig. 2. **D2&D3: Prediction concentration under the original input, subject removal, and context replacement.** Top1 label means the most frequently predicted label; Top2/3 labels denote the second- and third-most frequently predicted labels. Greater concentration in Top1/3 indicates stronger prediction collapse.

partial alignment with human affective ambiguity. AffectGPT instead produces a negative correlation of -0.245, consistent with its behavior in D1.

A positive ρ_{all} could arise simply because models are confident when correct and uncertain when incorrect. However, ρ_{correct} remains positive for the same four models, while AffectGPT remains close to zero. Emotion-LLaMA even increases from 0.365 to 0.541 on correctly predicted samples. Partial correlations controlling for prediction correctness are also significantly positive, showing that the observed alignment cannot be explained solely by correctness.

Rank alignment, however, does not imply calibrated confidence. Overall, the strongest models are reasonably calibrated (ECE 0.057 for Qwen3.5-9B; Table IV). We divide samples into high and low human agreement groups ($h_{\text{human}} \geq 0.8$ and $h_{\text{human}} \leq 0.5$, respectively) The high agreement threshold corresponds to the median of 0.800, while the low agreement threshold identifies samples without a clear majority. The confidence gap $\bar{p}_{\text{top1}} - \text{Acc}$ is small in the high agreement group (at most $+0.11$), whereas in the low agreement group accuracy drops sharply but p_{top1} does not, leaving gaps of $+0.24$ to $+0.43$ for the four positively correlated models. Cross-validated temperature scaling still leaves gaps of $+0.17$ to $+0.21$ on this group, so overconfidence is not a global temperature effect (SM A.4). Emotion-LLaMA has the lowest overall p_{top1} of 0.547, while AffectGPT is more confident on the low agreement group.

Figure 4 further compares model and human entropy. For the three general-purpose models, many samples with high human entropy fall below the diagonal, indicating lower model uncertainty when human judgments are highly am-

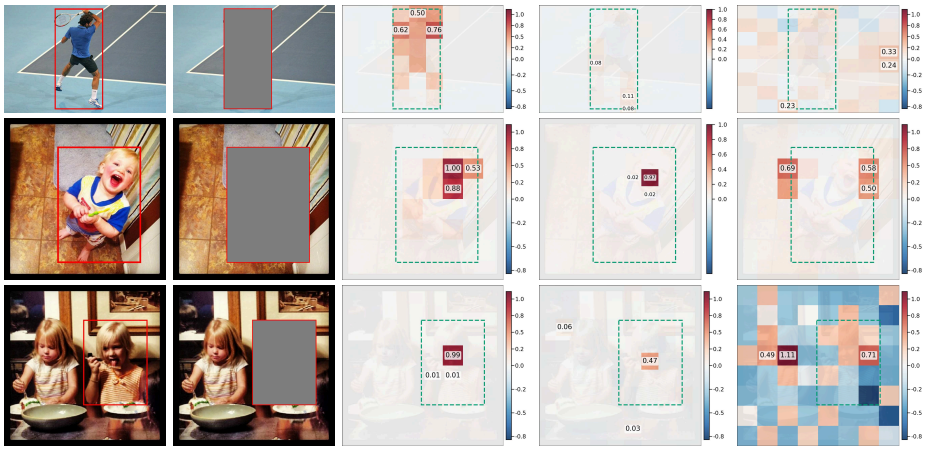


Fig. 3. **Case Study: PRMs for three EMOTIC samples at the 25% depth of each VLM's decoder.** Each row shows the original and subject-masked inputs followed by PRMs from three VLMs. The three rightmost columns show PRMs from different VLMs. The manually drawn green dashed rectangle marks the subject region. Higher PRM values indicate a larger increase in the probability of the full-input prediction after restoring that visual-token state. Maps are model-specific due to different visual token grids.

TABLE IV

D4: ACCURACY, CONFIDENCE, AND CALIBRATION ACROSS HUMAN AGREEMENT GROUPS. H AND L DENOTE HIGH ($h_{\text{human}} \geq 0.8$) AND LOW ($h_{\text{human}} \leq 0.5$) AGREEMENT, RESPECTIVELY. GAP $= \bar{p}_{\text{top1}} - \text{Acc}$ (> 0: OVERCONFIDENT); TS: 5-FOLD CROSS-VALIDATED TEMPERATURE SCALING.

Model	Acc. H	Acc. L	p_{top1} H	p_{top1} L	ECE	Brier	JSD	Gap L (→TS)
Qwen3.5-9B	0.936	0.472	0.923	0.723	0.057	0.297	0.130	+0.251→+0.181
InternVL3.5-8B	0.877	0.371	0.905	0.739	0.132	0.431	0.202	+0.369→+0.178
Ministral-3-8B	0.743	0.356	0.855	0.786	0.215	0.619	0.290	+0.429→+0.170
Emotion-LLaMA	0.608	0.238	0.620	0.475	0.106	0.629	0.364	+0.236→+0.205
AffectGPT	0.329	0.393	0.989	0.996	0.633	1.263	0.592	+0.603→+0.028

biguous. Emotion-LLaMA exhibits broadly higher entropy, including on samples with low human entropy, while AffectGPT remains concentrated at low model entropy. Overall, the models capture a coarse ranking of sample difficulty without fully matching the level of human affective ambiguity. Further discussion is provided in SM B.

Key Finding 4. Model uncertainty generally increases with human disagreement, but models remain overconfident on human-ambiguous images, indicating only partial alignment with human affective ambiguity.

V. DISCUSSION AND CONCLUSION

We present the first systematic diagnosis of MIRAGE-like behavior in visual emotion recognition, revealing label-prior collapse without spontaneous abstention, person-dominant grounding under intervention, and uncertainty alignment that breaks down on human-ambiguous images.

Future work can extend the diagnosis to additional evidence sources such as background, illumination, and other contextual cues. The basic prompt requires an emotion label, so the high no-vision mirage rate should be interpreted under this output constraint. Explicitly offering abstention mitigates this behavior for some models, but can also increase false abstention on real images.

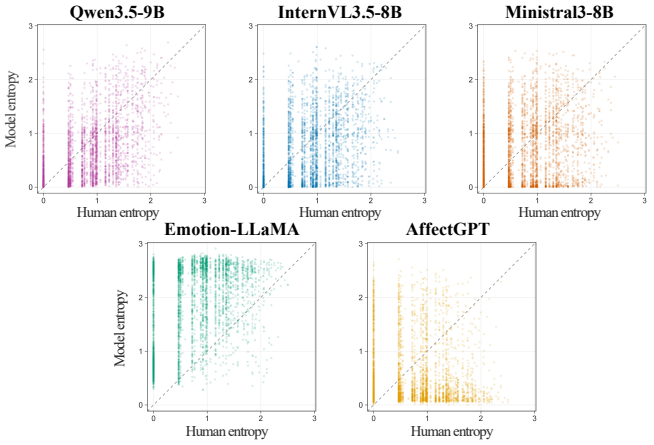


Fig. 4. **D4: Comparison of model and human entropy.** Each point represents one sample. Emotion-LLaMA exhibits generally higher uncertainty, whereas the other models show lower entropy.

Our findings suggest three practical recommendations. First, affective VLMs should be evaluated under missing or uninformative visual inputs with an explicit abstention option, reporting false abstention on real inputs and prediction distributions alongside accuracy. Second, emotion-specialized models should be tested across datasets, label spaces, and evidence conditions to assess whether specialization improves grounding rather than merely reinforcing label priors. Third, benchmarks should retain annotator vote distributions and report calibration conditioned on human agreement. Together, these diagnoses expose limitations obscured by benchmark accuracy and provide guidance toward more grounded and reliable affective VLMs.

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